

AIM : To estimate software development effort, time, and team size for the Attendance Planning and Analytics System using the COCOMO model.

```
1  """
2  Basic COCOMO (Constructive Cost Model) Calculator
3  Attendance Planning and Analytics System
4
5  Computes:
6      Effort (E)          = a * (KLOC)^b          person-months
7      Development Time (D) = c * (E)^d            months
8      Average Staff (P) = E / D                  persons
9
10 Modes:
11     Organic      -> a=2.4, b=1.05, c=2.5, d=0.38
12     Semi-detached -> a=3.0, b=1.12, c=2.5, d=0.35
13     Embedded     -> a=3.6, b=1.20, c=2.5, d=0.32
14 """
15
16 # Mode -> (a, b, c, d) coefficients
17 COCOMO_MODES = {
18     "organic":      {"a": 2.4, "b": 1.05, "c": 2.5, "d": 0.38},
19     "semi-detached": {"a": 3.0, "b": 1.12, "c": 2.5, "d": 0.35},
20     "embedded":     {"a": 3.6, "b": 1.20, "c": 2.5, "d": 0.32},
21 }
```

```
33
34 v def calculate_cocomo(kloc: float, mode: str):
35 v     """
36     Calculate Effort, Development Time, and Average Staff
37     using the Basic COCOMO model.
38
39     Args:
40         kloc (float): Estimated size of the project in KLOC (thousands of
lines of code)
41         mode (str): One of "organic", "semi-detached", "embedded"
42
43     Returns:
44         dict: effort, time, staff, and the coefficients used
45     """
46     mode = mode.strip().lower()
47 v     if mode not in COCOMO_MODES:
48         raise ValueError(
49             f"Invalid mode '{mode}'. Choose from: {'',
50             f'.'.join(COCOMO_MODES.keys())}"
51         )
52     if kloc <= 0:
53         raise ValueError("KLOC must be a positive number.")
54
55     coeffs = COCOMO_MODES[mode]
56     a, b, c, d = coeffs["a"], coeffs["b"], coeffs["c"], coeffs["d"]
```

```
67     effort = a * (kloc ** b)           # person-months
68     time = c * (effort ** d)           # months
69     staff = effort / time              # average number of people
70
71     return {
72         "mode": mode,
73         "kloc": kloc,
74         "coefficients": coeffs,
75         "effort_person_months": round(effort, 2),
76         "development_time_months": round(time, 2),
77         "average_staff": round(staff, 2),
78     }
79
80
81 def print_report(result: dict):
82     """Pretty-print the COCOMO result."""
83     print("\n" + "=" * 45)
84     print(" BASIC COCOMO ESTIMATION REPORT")
85     print("=" * 45)
86     print(f" Project Size (KLOC) : {result['kloc']}")
87     print(f" Mode Selected       : {result['mode'].title()}")
88     c = result["coefficients"]
89     print(f" Coefficients          : a={c['a']}, b={c['b']}, c={c['c']}, d=
90 {c['d']}")
91     print("-" * 45)
92     print(f" Effort (E)              : {result['effort_person_months']} person-
```

```
71     print(f" Effort (E)           : {result['effort_person_months']} person-  
      months")  
72     print(f" Development Time (D): {result['development_time_months']} months")  
73     print(f" Average Staff (P)    : {result['average_staff']} persons")  
74     print("=" * 45 + "\n")  
75  
76  
77     def run_interactive():  
78         """Interactive CLI: user enters KLOC and selects a mode."""  
79         print("Basic COCOMO Calculator – Attendance Planning and Analytics System")  
80         print("Available modes: Organic | Semi-detached | Embedded\n")  
81  
82         try:  
83             kloc = float(input("Enter estimated size in KLOC (e.g. 5): ").strip())  
84             mode = input("Select mode (organic / semi-detached / embedded):  
      ").strip())  
85             result = calculate_cocomo(kloc, mode)  
86             print_report(result)  
87         except ValueError as e:  
88             print(f"\nError: {e}\n")  
89  
90  
91     if __name__ == "__main__":  
92         # ---- Option 1: Interactive mode (uncomment to use) ----  
93         run_interactive()  
94
```

OUTPUT :

```
Basic COCOMO Calculator – Attendance Planning and Analytics System
Available modes: Organic | Semi-detached | Embedded
```

```
Enter estimated size in KLOC (e.g. 5): 30
```

```
Select mode (organic / semi-detached / embedded): organic
```

```
=====
BASIC COCOMO ESTIMATION REPORT
=====
Project Size (KLOC) : 30.0
Mode Selected       : Organic
Coefficients        : a=2.4, b=1.05, c=2.5, d=0.38
-----
Effort (E)          : 85.35 person-months
Development Time (D): 13.55 months
Average Staff (P)   : 6.3 persons
=====
```

```
Basic COCOMO Calculator – Attendance Planning and Analytics System
Available modes: Organic | Semi-detached | Embedded
```

```
Enter estimated size in KLOC (e.g. 5): 30
```

```
Select mode (organic / semi-detached / embedded): semi-detached
```

```
=====
BASIC COCOMO ESTIMATION REPORT
=====
Project Size (KLOC) : 30.0
Mode Selected       : Semi-Detached
Coefficients        : a=3.0, b=1.12, c=2.5, d=0.35
-----
Effort (E)          : 135.36 person-months
Development Time (D): 13.93 months
Average Staff (P)   : 9.72 persons
=====
```

```
Basic COCOMO Calculator – Attendance Planning and Analytics System
Available modes: Organic | Semi-detached | Embedded
```

```
Enter estimated size in KLOC (e.g. 5): 30
```

```
Select mode (organic / semi-detached / embedded): embedded
```

```
=====
```

```
BASIC COCOMO ESTIMATION REPORT
```

```
=====
```

```
Project Size (KLOC) : 30.0
```

```
Mode Selected       : Embedded
```

```
Coefficients        : a=3.6, b=1.2, c=2.5, d=0.32
```

```
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```

```
Effort (E)          : 213.23 person-months
```

```
Development Time (D): 13.91 months
```

```
Average Staff (P)   : 15.33 persons
```

```
=====
```